QMS Journal Club

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QMS Journal Club 5th meeting: Density matrix formalism and reduced density matrices

Previously, we mentioned that density matrix formalism is useful when

1. the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
2. when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).

Consider the Bell state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{1}$$

The description of this Bell state is that when measurement is performed for qubit A, there is a probability $\frac{1}{2}$ to get $|0\rangle$ or $|1\rangle$.

How can we describe the quantum state for qubit A alone?

One might be tempted to write

$$|\psi_A\rangle = \frac{1}{\sqrt{2}}(|0_A\rangle + |1_A\rangle). \tag{2}$$

This description gives you the correct statistics when you measure in the Pauli Z basis, but it gives you a wrong statistics when you measure in the Pauli X basis because $|\psi_A\rangle$ is now its eigenstate, $|+\rangle$.

This again shows that the state vector formalism does not completely capture the classical statistical ensemble aspect of mixed states.

Previously, an example of maximally mixed state was given,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3}$$

Regardless of what measurement basis we are taking, we always get a probability of $\frac{1}{2}$ to get either $\{|R\rangle, |L\rangle\}$ (or $\{|H\rangle, |V\rangle\}$).

Postulates of quantum mechanics

Postulate 1: Associate to any isolated physical system is a complex vector space with inner product, known as the state space of the system. The system is completely described by its density operator, which is a positive semi-definite operator $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with trace one, acting on the state space of the system.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. The state ρ of the system at time t_1 is related to the state ρ' of the system at time t_2 by a unitary operator U,

$$\rho' = U\rho U^\dagger. \tag{4}$$

Postulate 3: Quantum measurements are described by a collection of measurement operators M_m, such that the probability that result m occurs for the state of a quantum system ρ is given by

$$p(m) = Tr(M_m^\dagger M_m \rho), \tag{5}$$

and the state of the system after the measurement is

$$\frac{M_m \rho M_m^\dagger}{Tr(M_m^\dagger M_m \rho)}. \tag{6}$$

For projective measurement $M = \sum_m m|m\rangle\langle m|$ with projector $P_m = |m\rangle\langle m|$, $P_m^\dagger P_m = P_m$ and hence

$$\begin{aligned} p(m) &= \sum_i p_i \langle\psi_i|P_m|\psi_i\rangle = \sum_i p_i \langle\psi_i|P_m \sum_j |j\rangle\langle j|\psi_i\rangle \\ &= \sum_{ij} p_i \langle j|\psi_i\rangle\langle\psi_i|P_m|j\rangle = Tr\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|P_m\right) \\ &= Tr(\rho P_m) = Tr(P_m \rho). \end{aligned} \tag{7}$$

The expectation value of a measurement M follows the same argument and can be written as

$$\langle M \rangle = Tr(M\rho). \tag{8}$$

Postulate 4: The state space of a composite quantum system is the tensor product of the state spaces of the component quantum systems, such that the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \ldots \otimes \rho_n$.
Suppose that a density matrix is given as

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|. \tag{9}$$

Let

$$|a\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle, \tag{10}$$

$$|b\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle. \tag{11}$$

Then,

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1| = \frac{1}{2}|a\rangle\langle a| + \frac{1}{2}|b\rangle\langle b|. \tag{12}$$

Different ensembles of quantum states can give rise to the same density matrix. When will two sets of state vectors $|\psi\rangle$ and $|\phi\rangle$ generate the same density matrix?

Theorem: The sets $|\psi_i\rangle$ and $|\phi_j\rangle$ generate the same density matrix if and only if

$$|\psi_i\rangle = \sum_j U_{ij}|\phi_j\rangle, \tag{13}$$

where U_{ij} is a unitary matrix.

Reduced density matrix

Suppose we have a quantum system A and B, defined by ρ_{AB}. The reduced density operator for system A is defined by

$$\rho_A = Tr_B(\rho_{AB}), \tag{14}$$

where Tr_B is called partial trace over system B. Partial trace is defined by

$$Tr_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| Tr(|b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2|\langle b_2|b_1\rangle. \tag{15}$$

Question 1: Consider the Bell state $|\psi_{AB}\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$. Find the reduced density matrix of the second qubit.

Question 2: Suppose a composite quantum system A and B is in the state $|a\rangle|b\rangle$. Show that the reduced density matrix of system A is a pure state.

Quantum teleportation

Quantum teleportation begins with two parties, Alice and Bob, sharing a pair of entangled qubits in one of the Bell bases,

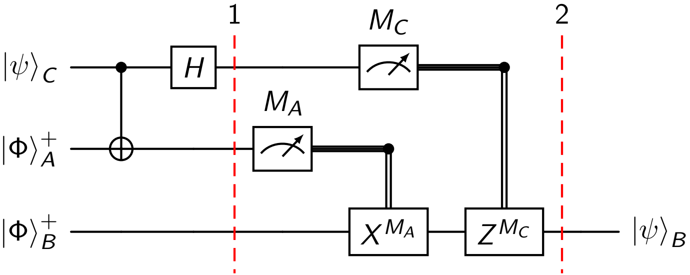
$$|\Phi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{16}$$

$$|\Phi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{17}$$

$$|\Psi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{18}$$

$$|\Psi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{19}$$

Without loss of generality, we focus on the Bell basis $|\Phi\rangle_{AB}^{+}$ in our discussion. The other Bell bases follow the same mathematical arguments.


1
2
$|\psi\rangle_C$
$|\Phi\rangle^+_A$
$|\Phi\rangle^+_B$
H
M_C
M_A
X^{M_A}
Z^{M_C}
$|\psi\rangle_B$

Initially, the shared Bell state between Alice and Bob, and the qubit that Alice wants to send to Bob is given as:

$$\begin{aligned} |\psi\rangle_{CAB} &= (\alpha\,|0\rangle_C + \beta\,|1\rangle_C) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \\ &= \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|100\rangle + \beta\,|111\rangle). \end{aligned} \tag{20}$$

After the CNOT gate, the above quantum state becomes:

$$|\psi\rangle_{CAB} = \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|110\rangle + \beta\,|101\rangle). \tag{21}$$

After the Hadamard gate, the quantum state at Step 1 will become:

$$\begin{aligned}
|\psi\rangle_{CAB} &= \frac{1}{\sqrt{2}}\left[\frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |00\rangle_{AB} + \frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |11\rangle_{AB}\right. \\
&\qquad \left. + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |10\rangle_{AB} + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |01\rangle_{AB}\right] \\
&= \frac{1}{2}[\alpha\,|000\rangle + \alpha\,|100\rangle + \alpha\,|011\rangle + \alpha\,|111\rangle + \beta\,|010\rangle - \beta\,|110\rangle \\
&\quad + \beta\,|001\rangle - \beta\,|101\rangle] \\
&= \frac{1}{2}[|00\rangle_{CA}(\alpha\,|0\rangle_B + \beta\,|1\rangle_B) + |01\rangle_{CA}(\beta\,|0\rangle_B + \alpha\,|1\rangle_B) \\
&\quad + |10\rangle_{CA}(\alpha\,|0\rangle_B - \beta\,|1\rangle_B) + |11\rangle_{CA}(-\beta\,|0\rangle_B + \alpha\,|1\rangle_B)].
\end{aligned} \tag{22}$$

In Step 2, quantum measurements will be performed on Alice's qubits (qubit A and C) and the appropriate Pauli correction will be applied to Bob's qubit (qubit B) to recover the quantum information that is teleported. In short,

1. If the quantum measurement results in $|00\rangle_{CA}$, nothing ($\sigma_0 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$) will be performed;
2. If the quantum measurement results in $|01\rangle_{CA}$, Pauli X, $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ will be performed;
3. If the quantum measurement results in $|10\rangle_{CA}$, Pauli Z, $\sigma_3 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ will be performed;
4. If the quantum measurement results in $|11\rangle_{CA}$, Pauli Y up to a global phase of i, $i\sigma_2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ will be performed.

Find the density matrix of $|\psi\rangle_{CAB}$,

$$|\psi\rangle_{CAB} = \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\beta|0\rangle + \alpha|1\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(-\beta|0\rangle + \alpha|1\rangle)]. \tag{23}$$

Verify that when Alice's system is being traced out, the reduced density matrix of Bob is maximally mixed.

This shows that any measurements performed by Bob will contain no information about $|\psi\rangle_{CAB}$, and Alice cannot use quantum teleportation to transmit information to Bob faster than speed of light.

Here are some intermediate steps for your verifications. Let

$$\begin{aligned}
|B_1\rangle &= \alpha\,|0\rangle + \beta\,|1\rangle\,, \\
|B_2\rangle &= \beta\,|0\rangle + \alpha\,|1\rangle\,, \\
|B_3\rangle &= \alpha\,|0\rangle - \beta\,|1\rangle\,, \\
|B_4\rangle &= -\beta\,|0\rangle + \alpha\,|1\rangle\,.
\end{aligned}$$

Then

$$\begin{aligned}
\rho_{AB} = \frac{1}{4}[&|0\rangle\langle 0| \otimes |B_1\rangle\langle B_1| + |0\rangle\langle 1| \otimes |B_1\rangle\langle B_2| + |1\rangle\langle 0| \otimes |B_2\rangle\langle B_1| \\
&+ |1\rangle\langle 1| \otimes |B_2\rangle\langle B_2| + |0\rangle\langle 0| \otimes |B_3\rangle\langle B_3| + |0\rangle\langle 1| \otimes |B_3\rangle\langle B_4| \\
&+ |1\rangle\langle 0| \otimes |B_4\rangle\langle B_3| + |1\rangle\langle 1| \otimes |B_4\rangle\langle B_4|]. \qquad (24)
\end{aligned}$$

Lastly,

$$\begin{aligned}
\rho_B &= \frac{1}{4}[|B_1\rangle\langle B_1| + |B_2\rangle\langle B_2| + |B_3\rangle\langle B_3| + |B_4\rangle\langle B_4|] \\
&= \frac{1}{4}[2(|\alpha|^2 + |\beta|^2)|0\rangle\langle 0| + 2(|\alpha|^2 + |\beta|^2)|1\rangle\langle 1|] = \frac{1}{2}I. \qquad (25)
\end{aligned}$$